

Monte Carlo Results for the Two-Dimensional Random-Bond Ising Model

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1 Model and the Griffiths region

We consider spins $s_i = \pm 1$ on an $L \times L$ square lattice with periodic boundary conditions and $N = L^2$ sites. Setting $k_B = 1$, the zero-field Hamiltonian is

$$H = - \sum_{\langle ij \rangle} J_{ij} s_i s_j, \quad J_{ij} = \begin{cases} 1, & \text{with probability } p, \\ 0, & \text{with probability } 1 - p. \end{cases} \quad (1)$$

Bonds are drawn independently and remain fixed during thermal sampling. This is the bond-diluted limit of the ferromagnetic random-bond Ising model. Dataset `run_005` uses $p = 0.6$. We denote thermal averages in realization a by $\langle \cdot \rangle_a$ and disorder averages by $[\cdot]_{\text{dis}}$.

Disorder allows unusually well-connected, nearly pure regions inside a globally paramagnetic system. In the Griffiths region, $T_c(p) < T < T_G$, these rare regions can generate enhanced responses and slow relaxation, while the equilibrium free energy is nonanalytic in magnetic field at zero field. Here $T_G = 2/\ln(1 + \sqrt{2}) \simeq 2.269$ is the pure square-lattice transition temperature. Classical Griffiths singularities can be weak: slow dynamics or broad response distributions alone do not establish their asymptotic form.

The present concentration lies above the square-lattice bond-percolation threshold $p_c = 1/2$. A percolating occupied-bond network and a nonzero thermodynamic ordering temperature $T_c(0.6)$ are therefore possible and, for this model, expected. The scan covers both low-temperature ordering behavior and the paramagnetic regime. We do not determine $T_c(0.6)$ from a single heat-capacity maximum or classify every temperature below T_G as part of the Griffiths region.

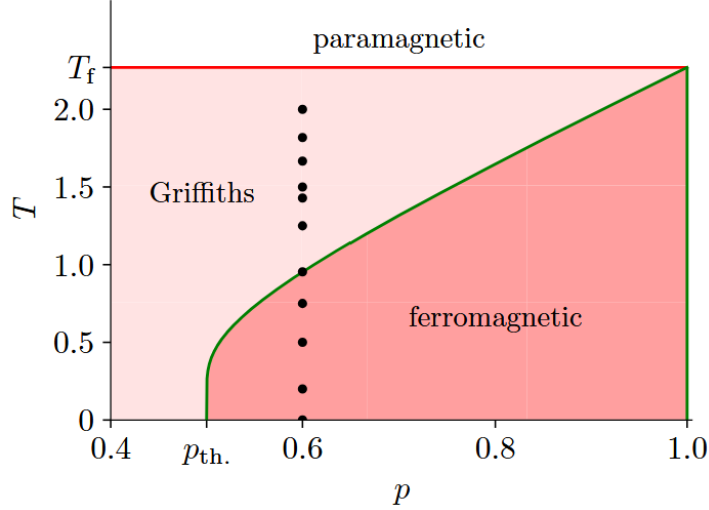


Figure 1: Schematic temperature–bond-concentration phase diagram. The green curve denotes $T_c(p)$, separating ferromagnetic order from the paramagnetic phase. The red line is the pure-system transition temperature, labeled T_f in the image and denoted T_G here. The lightly shaded region between these boundaries is the Griffiths region. The label $p_{\text{th.}}$ corresponds to the percolation threshold $p_c = 1/2$. Black markers illustrate a scan at the present concentration $p = 0.6$; their temperatures are schematic rather than the complete simulation grid.

Figure 1 emphasizes that the Griffiths region is globally paramagnetic despite its slowly relaxing, locally correlated rare regions. At fixed $p = 0.6$, cooling crosses T_G before reaching $T_c(0.6)$. Below the latter temperature, bulk order also contributes to large responses and persistent memory. The green curve is schematic and is not a fit to this dataset; a quantitative phase assignment requires an independent finite-size analysis of the ordering transition.

2 Monte Carlo procedure

The simulations used SpinMonteCarlo.jl 1.2.2 with Julia 1.13.0. Each realization was thermalized with 2048 Swendsen–Wang (SW) updates: parallel neighboring spins are linked with probability $1 - e^{-2J_{ij}/T}$, and each resulting cluster is independently assigned either orientation. Static observables were then measured after each of 16384 additional SW updates. Local responses used the same SW thermalization and measurement procedure. For the first disorder realization, the local-response calculation reused its MC seed and reproduced the static measurement trajectory.

Dynamical measurements were performed separately. After the last SW measurement, its final spin configuration was saved as the reference state and evolved for 4096 random-site heat-bath sweeps. Each sweep selects N sites with replacement and resamples them according to

$$P(s_i = +1 \mid \{s_j\}) = \frac{1 + \tanh(h_i/T)}{2}, \quad h_i = \sum_{j:\langle ij \rangle} J_{ij}s_j. \quad (2)$$

These heat-bath sweeps contribute only to the dynamical overlap, not to static averages. This separation permits efficient equilibrium sampling while retaining a specified local stochastic dynamics; SW update counts are not interpreted as physical relaxation times.

The sizes are $L = 10, 16, 24, 32, 40, 50$, with $T = 0.02, 0.04, \dots, 3.00$ and 1000 disorder realizations per size, giving 150 temperatures. The same bond realizations are reused across temperatures, with temperature-dependent MC seeds. The master seed is 1234. Static measurements are divided into 64 blocks of 256 SW updates for block statistics and jackknife estimates. Disorder-average bands use the sample standard deviation divided by $\sqrt{1000}$; this scatter already includes MC noise, so stored thermal errors are not added again. SW sampling improves cluster reorientation, but convergence still requires checks of run length, block size, and independent seeds.

3 Results

3.1 Disorder-averaged heat capacity and susceptibility

For $M = \sum_i s_i$, the observables are

$$C_a = \frac{\langle H^2 \rangle_a - \langle H \rangle_a^2}{T^2}, \quad \chi_a = \frac{\langle M^2 \rangle_a}{T}. \quad (3)$$

Both are total-system quantities; division by N gives per-site values. The susceptibility uses the exact symmetry $\langle M \rangle_a = 0$ of the finite-volume zero-field ensemble, without subtracting $\langle |M| \rangle_a^2$.

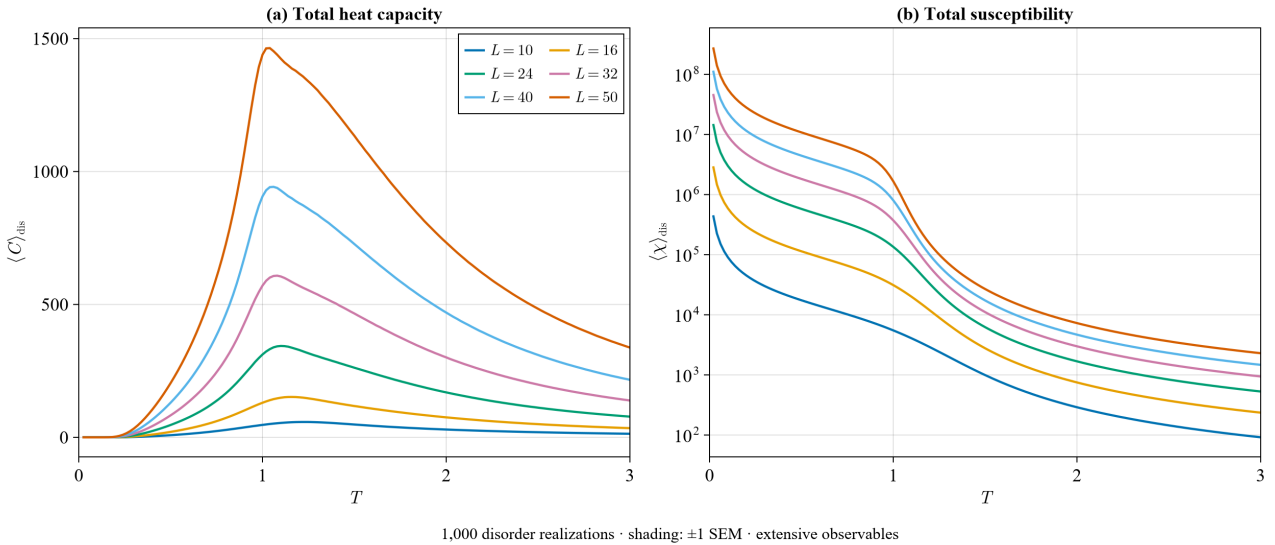


Figure 2: Disorder-averaged total heat capacity (left) and total susceptibility (right) for six sizes in `run_005`, with $p = 0.6$. Shading denotes ± 1 standard error of the disorder mean (SEM). The susceptibility axis is logarithmic.

The heat-capacity maximum shifts systematically toward lower temperatures as L increases. Its sampled location is $T = 1.24, 1.16, 1.10, 1.08, 1.06, 1.04$ for the six sizes, respectively. For $L = 50$, $[C]_{\text{dis}}/N \simeq 0.586$ at the maximum. The per-site peak varies only moderately, from 0.578 at $L = 10$ through 0.597 at $L = 24$ to 0.586 at $L = 50$. Thus much of the growth of the total peak reflects volume, whereas the peak displacement and sharpening contain finite-size information. Neither the peak height nor its location alone provides a thermodynamic transition temperature. In particular, the selected plotting temperature $T = 1.02$ is close to, but not equal to, the $L = 50$ heat-capacity maximum.

The susceptibility rises strongly upon cooling, without an isolated maximum in the displayed range. For this symmetric-ensemble definition, a large low-temperature value can include the squared magnetization of a macroscopically correlated cluster. It should not be interpreted solely as the connected response within one selected ordered phase. At high temperature, the ordinary extensive limit is $\chi \sim N/T$; separation between total-response curves alone does not demonstrate anomalous scaling.

For a fixed finite bond realization, the low-temperature equilibrium limit is particularly useful: $T\chi \rightarrow \sum_c n_c^2$, where n_c is the size of each connected occupied-bond cluster and its orientation is independently sampled. Reconstructing the 1000 bond realizations at $L = 50$ gives $[\sum_c n_c^2]_{\text{dis}} = 5.62517 \times 10^6$. At $T = 0.02$, the measured $[T\chi]_{\text{dis}}$ agrees with this value to better than 0.01%. The samplewise ratios $T\chi_a / \sum_c n_{c,a}^2$ have 5th and 95th percentiles 0.999775 and 1.000188. This supports the low-temperature SW sampling, without establishing convergence at every temperature.

3.2 Spatial structure of the local susceptibility

The response of an individual spin to a uniform field is

$$\chi_{i,a} = \frac{\langle s_i M \rangle_a}{T} = \frac{1}{T} \sum_j \langle s_i s_j \rangle_a, \quad \sum_i \chi_{i,a} = \chi_a. \quad (4)$$

Figure 3 follows the first $L = 50$ disorder realization across four temperatures, so the bond geometry remains fixed.

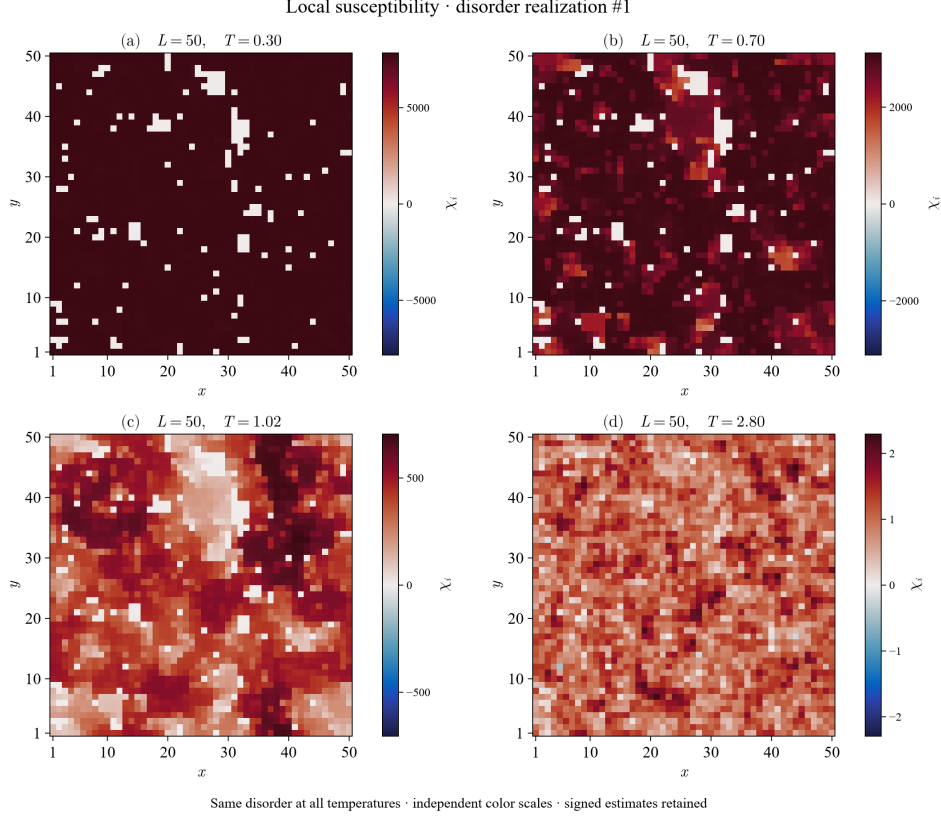


Figure 3: SW estimates of local susceptibility at $T = 0.30, 0.70, 1.02, 2.80$ for the same bond realization at $p = 0.6$. Each panel uses an independent symmetric linear color scale. Negative estimates are retained; equal color intensity across panels does not mean equal response.

At $T = 0.30$, most of the map has a large, nearly uniform positive response, interspersed with small weak-response regions. This is consistent with a dominant connected cluster: in the rigid-cluster limit, a site in a cluster of size n_c has $\chi_i = n_c/T$. At $T = 0.70$, variations within the strong-response region become visible, and at $T = 1.02$ the response forms a more heterogeneous pattern. At $T = 2.80 > T_G$, the map samples the high-temperature paramagnetic regime above the Griffiths region, and the overall response scale is much smaller. The maximum estimated response falls from approximately 7845 at $T = 0.30$ to 3119, 705, and 2.29 in the remaining panels. These maps show connectivity-dependent response, but the low-temperature large cluster is not by itself a rare region inside a paramagnetic background.

Ferromagnetic equilibrium correlations imply $\chi_{i,a} \geq 1/T$, whereas a finite-sample estimator can be negative. In the four panels, 2.40%, 2.36%, 2.16%, and 0.64% of sites have negative estimates, respectively; none lies below minus three times its stored block SEM. A weakly coupled or isolated spin can have a noisy $s_i M$ estimator because the instantaneous total magnetization is large, even when its own mean response is small. The small negative estimates therefore do not establish negative equilibrium response or antiferromagnetic patches. Their reported errors still require block-length checks, and sitewise tests are spatially correlated. The summed local estimates agree with the first

sample's total susceptibility at all selected sizes and temperatures, as expected from the shared SW trajectory.

3.3 Sample-to-sample susceptibility distributions

Figure 4 shows kernel density estimates of $x = \log_{10} \chi$ from 1000 realizations at each size and temperature. Their normalization is

$$\int p_x(x) dx = 1, \quad p_\chi(\chi) = \frac{p_x(\log_{10} \chi)}{\chi \ln 10}. \quad (5)$$

The ordinate is a density per unit logarithmic susceptibility, not per unit χ .

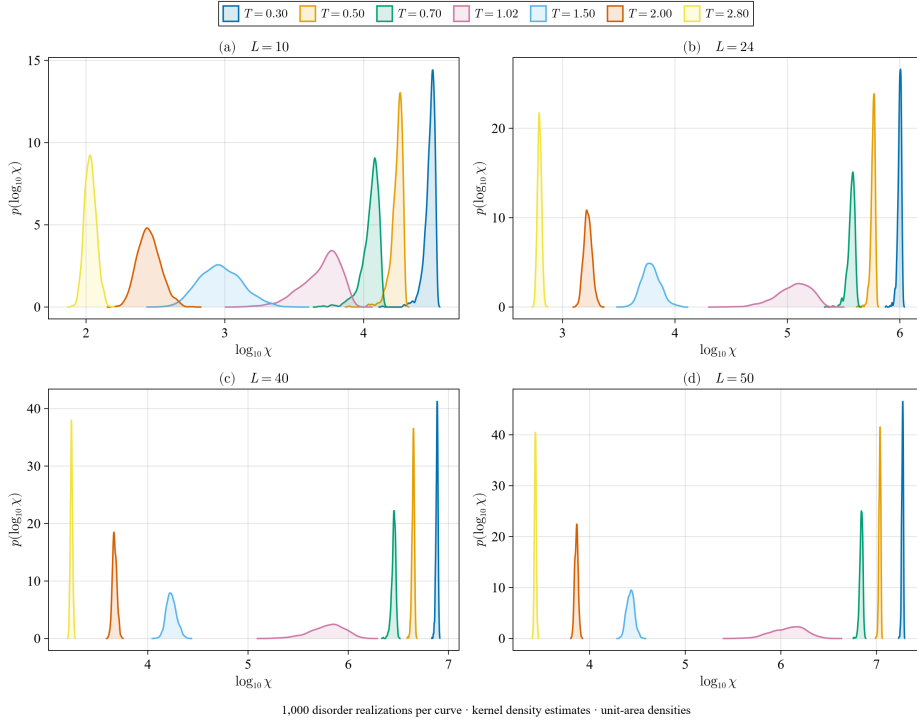


Figure 4: Distributions of total susceptibility at seven temperatures, $T = 0.30, 0.50, 0.70, 1.02, 1.50, 2.00, 2.80$, for $L = 10, 24, 40, 50$, normalized in $d(\log_{10} \chi)$. Smooth curves are kernel density estimates; narrow peaks have larger density because their areas remain unity. Small shoulders require bandwidth and sampling checks before interpretation.

Cooling shifts the distributions toward larger susceptibility, consistent with Fig. 2. For the larger sizes, the low-temperature curves at $T = 0.30, 0.50, 0.70$ are sharply concentrated, whereas $T = 1.02$ produces the broadest distribution among the seven plotted temperatures. At $L = 50$, the standard deviations of $\log_{10} \chi$ are 0.0085, 0.0096, 0.0159, 0.1674, 0.0398, 0.0177, 0.0095 in increasing temperature order. The broad intermediate-temperature response is compatible with sample-dependent thermal disruption of correlated clusters near the ordering crossover; it does not by itself locate T_c or establish a Griffiths tail.

The relative width $\text{std}(\chi)/[\chi]_{\text{dis}}$ at $T = 0.30$ decreases from 0.081 for $L = 10$ to 0.019 for $L = 50$. At low temperature and $p = 0.6$, a dominant occupied cluster can give a large response with relatively small sample-to-sample fluctuations. At $T = 1.02$, the relative width instead increases from 0.281 to 0.369 over these sizes. This fixed-temperature trend motivates a finite-size study but is not an asymptotic self-averaging test at a known critical point. At $T = 1.50, 2.00$, and 2.80 , the distributions narrow with increasing size. The added $T = 2.80 > T_G$ curve provides a high-temperature paramagnetic comparison outside the Griffiths region; its logarithmic width decreases from 0.0417 at $L = 10$ to 0.0095 at $L = 50$.

The rightward displacement with size partly reflects the total-system normalization. Replacing χ by χ/N translates a logarithmic distribution by $-\log_{10} N$ without changing its width. The densities include both quenched disorder variation and residual MC noise; independent SW runs and block-

size checks remain necessary for quantitative tail analysis. No power-law fit, Griffiths exponent, or universal distribution is inferred from these kernel density estimates.

3.4 Autocorrelation under heat-bath dynamics

The reference state is the final SW configuration after thermalization and static measurement. Starting from this state, the measured heat-bath overlap is

$$A_a(t) = \frac{1}{N} \sum_i s_i(t_0 + t) s_i(t_0), \quad 0 \leq t \leq 4096. \quad (6)$$

We use $A(t)$ to distinguish it from heat capacity; the plot labels it $C(t)$. Here t counts heat-bath sweeps from the reference, not SW updates. Each sample contributes one fixed-origin trajectory, without time-origin averaging or subtraction of spin means.

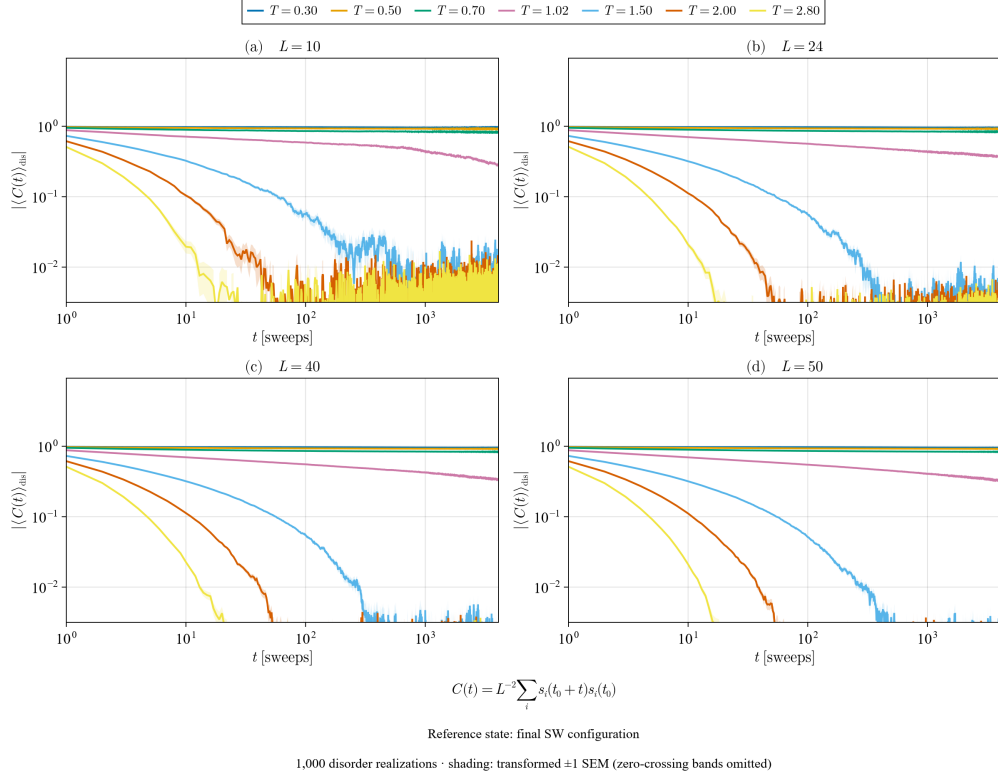


Figure 5: Absolute value of the disorder-averaged overlap, $|[A(t)]_{\text{dis}}|$, for the seven temperatures in Fig. 4. Both axes are logarithmic, with a vertical lower limit of $10^{-2.5}$. One time unit is one random-site heat-bath sweep following the last SW measurement. The absolute value is taken after disorder averaging. Shading is the absolute-value image of the original ± 1 SEM interval; intervals touching or crossing zero are omitted. The stored value $A(0) = 1$ is omitted, and exactly zero means are plotted as gaps. Values below the vertical limit are outside the displayed range.

The low-temperature curves show a rapid initial drop followed by persistent memory. At $L = 50$, $[A(4096)]_{\text{dis}} \simeq 0.955, 0.920, 0.834$ for $T = 0.30, 0.50, 0.70$, respectively, with relatively modest size dependence. Weakly constrained spins can relax quickly while large, strongly correlated regions retain their orientation. At $p > p_c$, such behavior can include low-temperature ferromagnetic correlations and must not automatically be assigned to the paramagnetic Griffiths region.

At $T = 1.02$, the $L = 50$ overlap decreases from 0.533 at 128 sweeps to 0.325 ± 0.004 at 4096 sweeps, resolving substantial relaxation beyond the initial window. The final value varies nonmonotonically with size, from 0.284 at $L = 10$ to 0.372 at $L = 24$ and 0.325 at $L = 50$. At $T = 1.50$ and 2.00, the $L = 50$ endpoint values are -0.00047 ± 0.00139 and 0.00022 ± 0.00095 , respectively; at $T = 2.80 > T_G$, the endpoint is 0.00070 ± 0.00076 . These signed values from the stored data are consistent with zero within the reported SEM, and their absolute values lie below the plotted lower limit $10^{-2.5} \simeq 0.00316$. Disappearance below this limit does not imply an exactly zero overlap.

These curves characterize local heat-bath dynamics, not the autocorrelation time of SW sampling. For a finite system at positive temperature, sufficiently long heat-bath evolution can ultimately reverse the magnetization; a finite-window plateau is not proof of a nonzero infinite-time limit. The log-log representation exposes early and late relaxation, but an apparently straight segment alone does not establish a power law. Taking the absolute value removes the sign of fluctuations about zero; late-time positive magnitudes therefore need not represent a resolved relaxation tail. The lower display limit also hides smaller magnitudes. Longer windows and independent reference trajectories are needed to distinguish decay forms. The SEM combines disorder variation with single-trajectory noise, and adjacent times remain correlated.

4 Summary

The four numerical figures describe static responses and local stochastic relaxation of the $p = 0.6$ bond-diluted Ising model. Above the occupied-bond percolation threshold, the low-temperature regime can support ferromagnetic order, whereas the Griffiths region is the paramagnetic interval between $T_c(0.6)$ and the pure-system temperature T_G . This distinction is necessary when interpreting both large susceptibility and long-lived memory.

The heat-capacity maximum moves from $T = 1.24$ to 1.04 as L grows from 10 to 50, while its per-site height changes comparatively little. This finite-size drift deserves further analysis, but the heat-capacity maximum alone does not determine the ordering temperature. The measured susceptibility uses the symmetric finite-volume ensemble and grows strongly upon cooling, with large connected clusters contributing through their squared magnetization.

SW updates are used for both thermalization and static measurements. A low-temperature check against the exact occupied-cluster expression $T\chi \rightarrow \sum_c n_c^2$ supports the $L = 50$ magnetic-response estimates. The local maps show a largely coherent low-temperature response that becomes spatially heterogeneous upon heating. Small negative local estimates remain possible because $s_i M$ fluctuates strongly; the displayed maps do not show large negative responses beyond three reported block standard errors. This is useful evidence about sampling quality, not a replacement for convergence tests across temperatures and observables.

The susceptibility distributions add information beyond their means. Low-temperature response is large but relatively narrowly distributed for the larger sizes, while the selected temperature $T = 1.02$ exhibits substantial sample-to-sample variation. At higher temperatures, including the added paramagnetic comparison at $T = 2.80 > T_G$, the distributions narrow with size. The response widths are consistent with changing connectivity and thermal correlation scales, but do not establish a universal tail, a lack of self-averaging, or a Griffiths exponent.

The subsequent heat-bath trajectories retain strong low-temperature memory over 4096 sweeps, show a slower decay near $T = 1.02$, and have endpoint estimates consistent with zero at $T = 1.50$, 2.00 , and 2.80 . Their absolute disorder averages are displayed on log-log axes with a lower limit of $10^{-2.5}$; this display does not resolve the sign or the sub-limit tail. Efficient SW sampling and slow local dynamics are compatible: they use different transition rules. Neither a finite-window plateau nor spatial heterogeneity alone identifies the rare-region contribution in the paramagnetic phase.

For quantitative conclusions, the next steps are to determine $T_c(0.6)$ through Binder crossings and finite-size consistency checks, verify static errors with longer runs and different block sizes, and extend dynamical sampling with independent reference states and longer observation windows. Separating equilibrium sampling, the specified heat-bath dynamics, and the location of the ordering transition is essential for a controlled investigation of Griffiths behavior.